

Designing an Autonomous Soil Monitoring Robot

Patrick M. Piper, Jacob S. Vogel, Matthew T. Fahrenkrug, Shannan J. McNamee,
Quang N. Pham, Gregory C. Lewin
University of Virginia, pmp5kh, jsv4ee, mtf2fa, sm8vu, qnp4gd, gcl8a@virginia.edu

Abstract - Through the monitoring of soil conditions land managers can respond rapidly to mitigate adverse events, such as extreme weather or ongoing drought. However, without an extensive system of sensors, gathering information over a large field takes an exorbitant amount of time. This mass collection of soil data would allow farm managers to study time-lapsed trends and variables within a particular region to provide quick assessment of land conditions. Currently, the client uses a bulky handheld wireless soil sensor to measure moisture content and temperature. To take measurements, the client must walk to the coordinates of interest, clear the ground of vegetation, manually insert the probe into the ground, and log the reading. The team is designing an autonomous soil monitoring rover to expedite data collection and reduce labor. The rover will be able to autonomously navigate through a field several acres in size and avoid obstacles. It will gather data on soil moisture and temperature at a set of given waypoints and relay the information back to the farm manager. Constructed with a custom welded steel frame, the first rover prototype will be a four-wheeled vehicle with front wheel drive. The vehicle will be equipped with a Stevens Hydra Probe II mounted to a linear actuator. Navigation will be handled using a GPS and wheel encoders. When completed, the rover will allow the land manager to analyze trends between soil data and pasture health, providing an accurate snapshot of a field.

Index Terms – Soil data, Land monitoring, Rover, Autonomous robot, Autonomous navigation

INTRODUCTION

To keep pastures healthy and productive, farmers must implement a variety of land management techniques. Traditionally, land managers have used their own experience to predict the health of paddocks and make decisions based on these observations. However, this task is difficult and time consuming, especially on larger plots of land. Further complicating matters, farmers do not have a comprehensive understanding of how specific pasture maintenance approaches actually affect the health of the land.

Innisfree Village, located 30 minutes northwest of Charlottesville, has been experimenting with novel management techniques to improve the health of its fields. In order to aid with the assessment of land conditions, the team will be developing an autonomous land-based robot for Innisfree that can collect and store soil information. The data

collected by the robot can be processed and analyzed to aid in rapid assessment of pasture health by means of time lapse trends, or measurements before and after specific land treatments or weather events. This information will not only help Innisfree improve the condition of their land, but will also improve our knowledge of how soil surface conditions relate to overall pasture health.

NEEDS AND REQUIREMENTS

In order to accomplish its mission, the robot must be able to travel through rough pasture conditions including thick vegetation and mud. This means that the chassis must be weather resistant and tough. The chassis should be designed such that the performance can be optimized for varied terrain. Since the robot will be operating over large open areas, pinpoint steering is not necessary. However, the steering mechanism must be sturdy in order to minimize the drift associated with movements over long distances. Because the rapid assessment of field conditions is a key objective, the power system must be able to provide sufficient power for an entire mission. This requires the robot to move at least 1.5 mph on level ground and be able to continuously operate for two hours, allowing even the largest Innisfree field to be mapped in a single cycle. Based on pasture conditions at Innisfree, the robot must also be able to maintain control up inclines of 15 degrees. Although not a requirement, a suspension system may be implemented in order to improve power and steering capabilities.

The robot must also be able to navigate autonomously. Innisfree currently uses a data management system called MyObservatory to store and view information on land conditions. The user will be able to specify a field to be visited, and MyObservatory will generate a set of ordered GPS waypoints within the field. The robot needs to determine its own location relative to the given soil sampling locations and then embark. The robot will also need to sense local obstacles (such as rocks, trees, etc.) and avoid them. It is necessary for the location of the robot to be known with an accuracy of 10 feet. This accuracy ensures that trends stretching across an entire field can be accurately mapped.

Once the robot reaches the specified location in a set, it must prepare the soil surface for measurement. Since rocks, foliage, and standing water at the soil surface can skew soil readings, the area must be cleared. Once any obstacles have been removed, the probe must be inserted into the ground. The probe should consistently reach a specified depth and the delivery system should be able to sense if insertion has been unsuccessful. If the system does not detect any

problems, measurements should then be taken and recorded for processing and analysis.

After the data has been collected, the information needs to be uploaded to Innisfree's data management system. The input can be as simple as a .CSV file with the following columns: date, time, latitude, longitude, moisture reading and temperature reading. Additionally, there will be an error checking method to determine whether or not a reading should be flagged and inspected further. Error checking is concerned with measured moisture values or GPS location. This method should have a very high tolerance for outlying moisture measurement due to the natural variability of soil.

DESIGN DECISIONS AND JUSTIFICATIONS

While robots have been used in a wide variety of agricultural applications, the unique task of monitoring soil in pastureland has not been attempted in this method. Since this project hinges on the accurate measurement of soil conditions, it is important that the soil probe be accurate, reliable, and durable.

I. Probe Delivery System

Since Innisfree has been using the Stevens Hydra Probe, the probe itself is adopted and integrated into the robot design. According to the client, the Hydra Probe is the industry standard in soil monitoring and takes readings of desired metrics at the resolution needed [1]. The probe has four tines to measure moisture, a ground-contact temperature sensor, and is housed inside a cylindrical, hard plastic shell. The data can be read from the probe through an SDI-12 digital communication protocol.



FIGURE 1

STEVENS HYDRA PROBE II SOIL MOISTURE AND SALINITY SENSOR [2]

In order to determine how the probe would be inserted into the ground, the client calculated the force needed to drive the tines into different soils. Based on these estimates, the maximum force required for the system was determined to be approximately 80 pounds. After considering several probe delivery options, such as a gravity or spring assisted drop system and a pile driver, a linear actuator best met all of the requirements. An actuator allows rapid deployment of the probe, while still supplying the necessary force and stability to drive the probe through any anticipated ground conditions. A linear actuator equipped with feedback (position and current consumption) gives us several more advantages. Using the position feedback, it can be determined where the probe is relative to the chassis. The current feedback can be used to calculate how hard the

motor is pushing the probe. Along with a set of buttons surrounding and flush with the face of the probe, the system can ascertain when the tines are completely inserted and the Hydra Probe's temperature sensor is flush with the ground. After testing the capabilities of a smaller model, Pololu's 12-inch Concentric Linear Actuator with feedback met the set requirements. The device is sturdy and is capable of supplying 110 pounds of force with position feedback.

The probe and delivery device needed to be connected in a way that allowed for consistent and reliable insertion. A two-piece custom mount that could bolt onto the end of the actuator was designed and 3D printed for testing. The mount is to be machined from a block of aluminum during the final construction. The inside is conformed to the ridges of the probe shell, ensuring that the probe does not slip, even under extreme pressure. Sleeves for four contact sensors were incorporated into the probe mount, and the contact sensors themselves were aligned so that they make contact with the ground only when the soil probe is correctly inserted. In order to cut down on false readings, the four sensors were arranged around the edges of the probe

II. Chassis and Drive System

The primary considerations for the chassis were the materials used for the skeleton, the organization of components, and the type of drive system. The total weight should be minimized because more weight would slow the entire system down and require more power. However, due to the selected method of probe delivery, the weight of the vehicle must be at least 110 pounds, so that the actuator does not lift the robot off of the ground. Steel was chosen as the primary material due to its high yield strength and to allow for parts to be easily welded. The first design called for a box-like chassis made entirely of steel. The use of only steel would greatly surpass the weight minimum specifications, requiring larger, heavier motors. Therefore, both steel and aluminum materials are used in the design. Long hollow steel tubes were welded together to form a "skeleton" frame to provide structure, support, and the ability to add the probe delivery system and electronics in a modular manner. Aluminum plates and brackets are used in order to secure the motors and wheels to the frame using bolts. The aluminum provides adequate support, while the bolting offers the potential to adjust spacing and heights if needed.

For the wheel and motor assembly, addressing the tall and thick vegetation on the farm was a major challenge. The motors could be damaged if they were immediately attached at the axle of the wheels due to rough terrain conditions. In order to provide sufficient ground clearance, chain-driven wheels are used. The motor and chassis frame are elevated above the ground using long hollow aluminum columns with the wheels at the bottom. The aluminum columns enclose this chain system and prevent the surrounding vegetation from getting caught in it.

Another major decision revolved around whether to use three or four wheels. The largest advantages of a four wheel design are the large amount of stability and the ease of

weight distribution. A disadvantage with implementing a four-wheeled design is that the robot would need to incorporate a suspension system to address the case where uneven terrain causes one wheel to leave the ground. A three-wheeled vehicle has the advantage that all three wheels make contact with the ground at any given time. In the end, the two alternatives were combined into a system that implemented three points of contact with four total wheels. Two wheels, and their respective motors, are mounted in the front of the vehicle. The remaining two wheels are smaller, free-spinning, and mounted tightly together in the rear of the vehicle, similar to the wheel design of an airplane. This compromise circumvents the suspension issue while providing additional stability.

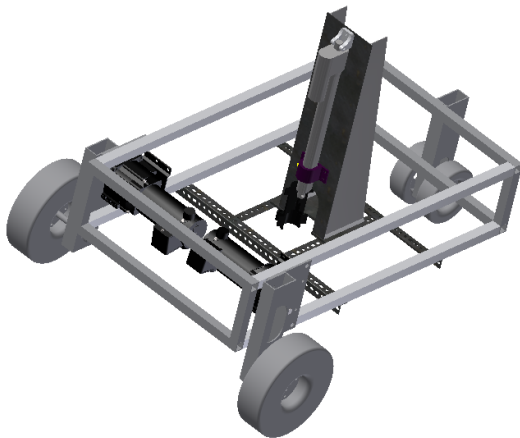


FIGURE 2
CAD ASSEMBLY OF 1ST GENERATION CHASSIS & PROBE

In order to make our robot fully autonomous, a steering system needed to be implemented. Original considerations included a traditional Ackerman steering system and individual steering for at least two wheels on the vehicle. Ackerman steering is the geometric arrangement of linkages between a pair of wheels in the steering of a car but has commonly been used on other types of vehicles. Individual wheel steering would require non-drive wheels to be motor-driven about an axis perpendicular to the ground. For this iteration of the project, a simple slip steering system was implemented instead of a more typical mechanical solution. This method involves changing the speeds of the two drive motors relative to one another in order to alter the direction of movement. By using slip steering, no additional mechanical parts were needed. While this method does not give the robot pinpoint steering, the system is accurate enough for testing purposes. Ackerman steering and individual wheel steering represent possible solutions to be implemented in future iterations.

Several requirements were considered in order to design a power system that best fit our purposes. First, it was determined that a fully electric system would provide the simplest and most efficient solution. Not only does this cut down on the components needed, but it also improves efficiency and allows for more precise control of the motors. Since mobile electronic power supplies, such as batteries,

typically operate at low direct current voltages, a low voltage DC motor was required. Maximum power and torque requirements were determined by calculating worst-case scenarios based on possibilities for the robot's weight, maximum grade percentages, and wheel size. In addition, the required maximum speed of the robot was estimated from earlier calculations. In order to achieve desired torque and speed characteristics, the DC motor needed to be geared down significantly. Ultimately, a $\frac{1}{2}$ horsepower 30:1 12V DC Gearmotor from Ironhorse Motors was selected. Two motors are to be mounted on the robot, one on each front drive wheel. Each motor is geared to spin freely at 56 rpm and has a torque of 168 in-lbs, which equates to a free robot speed of ~1.7 MPH. This meets the minimum requirements determined by our calculations. Since these motors draw 15.7 amps while under full load, a two channel motor driver was needed that could control at least 16 amps continuously and be able to handle significantly higher transient currents. A 30 amp (continuous) dual DC motor driver from Pololu met all of these specifications. An additional advantage to this driver is that it can be controlled with 5 V logic directly from the processor.

All of the current necessary to power both drive motors, as well as critical electronics, is sourced from a large battery. Because weight is not a critical concern, 12 V SLA (sealed lead acid) batteries were the best option. Typical SLA batteries can source several hundred amps if necessary. In addition, SLA batteries can be easily placed in parallel to increase the energy storage capacity of the robot. While other, more exotic batteries offer better weight to energy ratios, they are much more expensive, difficult to charge, and potentially dangerous. An 18 Ah SLA battery is currently being used in our tests, but a larger battery will be necessary for the final implementation.

III. Basic Navigation

The on-board navigation system needs to get the robot to within 10 feet of each desired waypoint. It is not necessary for the robot to be exactly on each waypoint due to the natural variability of soil. Further than 10 feet away will provide a skewed depiction of the land. The navigation system will combine wheel encoders and a GPS signal in an extended Kalman filter for its most basic level of navigation. An extended Kalman filter consolidates the position location from the two sources of information. Given the predicted location from the encoders and the measured location from the GPS, the filter produces a probabilistic prediction of the actual location based on noise parameters from each signal [3]. Noise from the encoders is the estimated error from slipping, and noise from the GPS is the estimated error in accuracy, drift, or large jumps.

The wheel encoders are two magnets fashioned 180 degrees apart on the shafts each drive motors. The signals will be read with a single Hall-effect sensor. Two sensors can be used to create a quadrature encoder, but is unnecessary because the direction is known from the power input to the motors. Two "ticks" per rotation results in a

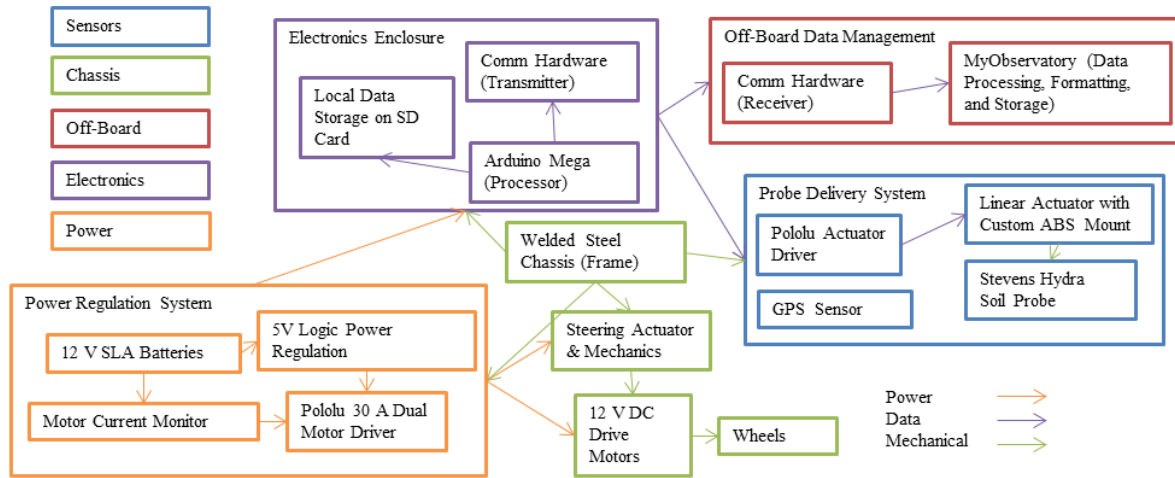


FIGURE 3
SYSTEM ARCHITECTURE

resolution of π times the radius of the wheels, which in the current design is less than 16 inches. This resolution is less than required to get close enough to the next waypoint. However, there will be a large amount of error from the encoders due to the slipping that occurs from rough terrain.

STATE OF CONSTRUCTION AND EVALUATION

After the assembly of the components, the first prototype is intended to drive 100m in a straight line and taking a soil reading every 10m. A successful completion of this task will demonstrate the robot's novel functions.

Initially, a probing prototype was constructed to take soil readings utilizing a linear actuator, a 3D printed mount, and a soil probe. Preliminary testing was completed using wooden mount that simulated how the soil device would be mounted on the chassis. The actuator was able to deploy and retract the probe in standard soil conditions and the probe successfully took soil readings. This met the functional requirements of applying adequate downward force, stable insertion, and accurate measurements. Currently, the linear actuator, 3D printed mount, and soil probe are attached to a custom-welded, steel tower that is mounted to the first generation chassis.

A robot chassis was designed and assembled via Autodesk. The model showed that the designed components would fit together once constructed. Stress simulations identified areas where the robot would experience the most stress and demonstrated that the chassis would not break or bend under high applied forces.

Initial navigation made use of only a GPS receiver on a stationary probe. To provide more accurate positioning encoders were added to the motors on the chassis and an extended Kalman filter was used to reconcile these data points. Initial testing in a confined environment was too limited to demonstrate the validity of the algorithm and proved a need for further testing in a larger scale area.

System components are still undergoing testing and construction.

FUTURE WORK

As the first functional prototype becomes operational, there are still a number of requirements that need to be fulfilled. The first generation chassis is a simple frame, with three ground contact points, and meant to be modular. The final iteration is expected to incorporate more advanced functional features as well as sleeker design. After the probe delivery and drive systems are successfully tested, excess chassis space can be utilized for additional features. The addition of a suspension system would allow the robot to traverse through the pasture more easily, as well as open up the option for a four contact point robot. Finally, in place of slip-steering method, a dynamic steering system should be implemented to allow for better navigation and obstacle avoidance.

Currently, the navigation system is a basic, custom system used for proof-of-concept. In order to improve the navigation system a magnetometer can be added in order to measure the robot's heading instead of calculating it based on a change in GPS location. One function that has not been addressed by the team at all is obstacle avoidance, a necessity for the robot to be truly autonomous. The tall tower used for the probe delivery system provides a good vantage point for sensors to be located. A laser scanner is a possible solution in this capacity [4].

Finally, the data management system should become wireless or tethered. The user should not have to manually upload the set of waypoints to the microcontroller, and then read data from an SD card. A wireless transmission of waypoints and soil data allows for the robot to be hands-off, which is preferable due to its autonomous nature. Real-time data transmission is also a possibility, but would require the

implementation of a telemetry system in the areas of interest, where cell phone signals are very poor.

CONCLUSION

After the first generation chassis demonstrates the robot's novel functions, further iterations can improve on physical design. The prototype chassis is designed in a way to have plenty of room for additions and changes. Developing a more permanent chassis should be the focus of further iterations. All of the robot's measurement and navigation functions are routed through a single microcontroller, which interprets information from the various sensors and transfers all of the collected data to an SD card. Ultimately, this prototype shows the practicality of accurate, reliable, and rapid collection of soil data through the deployment of an autonomous, wheeled robot. The completion of this device will to reduce manual labor and aid in the rapid assessment of pasture health.

REFERENCES

- [1] "Soil Monitoring Made Easy." 2011. <http://www.soilsensor.com/soilsensors.aspx>. Accessed: March 22, 2015
- [2] "Using the Stevens Hydra Probe with NI Wireless Sensor Networks (WSN)." 2011. <http://www.ni.com/product-documentation/9954/en/>. Accessed: March 26, 2015
- [3] Greg Welch and Gary Bishop. July 2006. An Introduction to the Kalman Filter. Department of Computer Science, University of North Carolina at Chapel Hill.
- [4] Abel Mendes, Luis Conde Bento, and Urbano Nunes. 2004. *Multi-target Detection and Tracking with a Laserscanner*. IEEE Intelligent Vehicles Symposium.

AUTHOR INFORMATION

Patrick M. Piper, Undergraduate Student, Dept. of Systems and Information Engineering, University of Virginia

Jacob S. Vogel, Undergraduate Student, Dept. of Electrical and Computer Engineering, University of Virginia

Matthew T. Fahrenkrug, Undergraduate Student, Dept. of Electrical and Computer Engineering, University of Virginia

Shannan J. McNamee, Undergraduate Student, Dept. of Mechanical and Aerospace Engineering, University of Virginia

Quang N. Pham, Undergraduate Student, Dept. of Mechanical and Aerospace Engineering, University of Virginia

Gregory C. Lewin, Lecturer, Dept. of Systems and Information Engineering, University of Virginia